0268. Missing Number

Easy · Bit Manipulation

leetcode.com/problems/missing-number

Problem

- Nums holds n distinct numbers from 0..n. Return the one that is missing.

Intuition

- XOR all indices 0..n with all values; every present number cancels with its index, leaving the missing one.
- The sum formula n(n+1)/2 - sum also works.

Brute Force

- Sort and scan (O(n log n)), or use a HashSet (O(n) space).

Optimal Approach

- X = n.
- For each i: x ^= i ^ nums[i].
- Return x.

Worked Example

```
[3,0,1], n = 3.

x = 3
i=0: x ^= 0 ^ 3 -> 0
i=1: x ^= 1 ^ 0 -> 1
i=2: x ^= 2 ^ 1 -> 2
answer 2
```

Java Solution

```java
public class MissingNumber {
    public int missingNumber(int[] nums) {
        int x = nums.length;
        for (int i = 0; i < nums.length; i++) x ^= i ^ nums[i];
        return x;
    }
}
```

Complexity

- Time: O(n). Space: O(1).

Pitfalls

- Start x at n, because the loop never visits index n.